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Quantifying Covariate Shift and Improving Electromyography Driven Gesture Recognition with Calibration and Sample Selection

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***Abstract*—Due to its non-stationary nature, surface electromyography (sEMG)-driven gesture recognition requires calibration to achieve satisfactory performance. Supervised domain adaptation is a commonly employed calibration technique. This technique involves combining one newly recorded repetition named calibration data of each gesture with all previously recorded data for each gesture under different conditions. This paper first quantified intra- and inter-conditions covariate shifts using the recognition error and Repeatability Index (RI), along with two newly introduced metrics: the *non-overlapping index* and the *non-overlapping cluster count*. Secondly, we extended the linear discriminant analysis (LDA) with the domain adaptation method by implementing a sample selection approach based on the repeatability index. We evaluated proposed techniques using two public datasets: the Multiday Ninapro DB6 and the Multiple Limb Position. Our domain adaptation approach with sample selection consistently outperformed the baseline model trained solely with calibration data in both datasets, showing improvements ranging from 0.7% to 3%. Compared to the domain adaptation without sample selection, our method demonstrated a 0.3% performance increase for the Multiple Limb Position dataset and achieved comparable results for the Ninapro DB6, underscoring the efficacy of RI-based sample selection in the domain adaptation.**

***Index Terms*—Domain Adaptation, Machine learning, Gesture Recognition, Surface Electromyography**

I. INTRODUCTION

SURFACE Electromyography (sEMG) is an emerging noninvasive sensory modality that has been the subject of thorough investigation in the fields of rehabilitation and prosthetic device control. Despite its significant potential, sEMG-driven applications still face major reliability challenges due to their non-stationary nature. For sEMG gesture recognition, supervised domain adaptation is a commonly used technique for sEMG calibration to mitigate performance degradation. During the sEMG calibration process, users are required to perform one repetition of each gesture in the predefined gesture set to form the calibration dataset. The main objective of the sEMG calibration process is to effectively leverage a small amount of newly recorded calibration dataset in combination with a larger existing training dataset to achieve improved overall performance.

Calibration of sEMGs gesture recognition has been the focus of numerous studies in recent times. Vidovic et al. [1] proposed domain adaptation with linear discriminant analysis to maintain gesture recognition performance over multiple days. Furthermore, numerous studies have investigated the potential of deep transfer learning in scenarios such as electrode shift [2], intersubject, and task transfer [3]. The principle of deep transfer learning is that neural weights, previously learned from a source dataset, contain valuable information that can be utilized for a new domain.

Many previous studies on sEMG calibration are limited to using simple training datasets, which are typically collected under a single condition. This approach contrasts with the established practice of recording sEMG gesture data under diverse conditions to address the nonstationary nature of sEMG [4-6]. For handling high-diversity datasets, Kim et al. [7] introduced a selective method using ensemble deep transfer learning for subject transfer learning. In their study, a pool of 10 models with the highest recognition rate on the calibration dataset was selected for fine tuning. Gestures were determined by the majority voting among the ensemble of models. Their proposed ensemble technique demonstrated a 3% improvement over a deep model pre-trained with data from 10 subjects and then fine-tuned with calibration data.

One major drawback of Kim et al. [7] approach is the use of an ensemble of deep neural networks, which is considerably computationally intensive. Additionally, due to the high complexity of deep neural networks, selecting support subjects based solely on recognition performance does not offer a detailed understanding of changes in the feature space. Bunderson and Kuiken [8] introduced three metrics to quantify changes in the feature space: Separability Index (SI), Repeatability Index (RI), and Mean Semi-principal Axis (MSA). These metrics are designed based on the intuition that underlies LDA, where recognition performance is based on two crucial factors: small within-class variance and large between-class variance.

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Inspired by Kim et al. [7], we extended the LDA domain adaptation of [1] using Repeatability Index (RI) based sample selection method. We evaluated our proposed method using both the Multiple Limb Position dataset [9] and Multiple Day Ninapro DB6 dataset [10] which respectively have medium to large covariate shifts. Moreover, building upon Bunderson and Kuiken's work [8], we have proposed two new metrics for quantifying covariate shift, namely the Non-Overlapping Index (NOI) and the Non-Overlapping Cluster Count (NCC).

Our paper is organized as follows: Section II provides an overview of the background material on machine learning algorithms, metrics for quantifying covariate shift, and our proposed domain adaptation approach for LDA with RI-based sample selection. Section III provides detailed information about the Khushaba Multiple Limb Position dataset [9] and multi-day Ninapro DB6 [10]. We present our findings in Section IV, followed by in-depth discussions in Section V. The paper concludes with our final remarks in Section VI.

II. METHODOLOGY

A. Fundamentals of Linear Discriminant Analysis

1) Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) is a popular technique in machine learning for sEMG gesture recognition [1]. Its widespread use is attributed to its parametric nature, simple linear decision boundary and most importantly, its computational efficiency making it suitable for real-time prosthetic control. Given a dataset represented by n column vectors of d dimensions $= \{\boldsymbol{x_1}, \cdots, \boldsymbol{x_n}\} \in R^{d \times n}$, where each sample $\boldsymbol{x_i}$ corresponds to one of c classes, LDA's main objective is to solve for subspace of size $(c-1) \leq d$ where the data is maximally separated among categories. This subspace can be obtained by solving the Rayleigh-Ritz quotient for the subspace $\boldsymbol{W} = \{\boldsymbol{w_1}, \cdots, \boldsymbol{w_n}\} \in R^{d \times (c-1)}$ [11]:

$$\arg\max_{\mathbf{W}} \frac{\mathrm{tr}(\mathbf{W}^\top \mathbf{S_B} \mathbf{W})}{\mathrm{tr}(\mathbf{W}^\top \mathbf{S_W} \mathbf{W})} \quad (1)$$

where $\mathrm{tr}(\cdot)$ represents the matrix trace operation, $\boldsymbol{S_B}$ and $\boldsymbol{S_W}$ are the between-class and within-class covariance matrices, respectively. Let μ denote the mean of the entire dataset, and μ_j of class j . The matrices $\boldsymbol{S_B}$ and $\boldsymbol{S_W}$ are calculated as follows:

$$\mathbf{S}_B \triangleq \sum_{j=1}^{c} (\boldsymbol{\mu}_j - \boldsymbol{\mu})(\boldsymbol{\mu}_j - \boldsymbol{\mu})^\top \quad (2)$$

$$\mathbf{S}_W \triangleq \sum_{j=1}^{c} \boldsymbol{S}_j = \sum_{j=1}^{c} \sum_{i=1}^{n_j} (\mathbf{x}_i^{(j)} - \boldsymbol{\mu}_j)(\mathbf{x}_i^{(j)} - \boldsymbol{\mu}_j)^\top \quad (3)$$

Optimization objective of Eq.1 can be approximated as:

$$\underset{\mathbf{W}}{\operatorname{argmax}}\, \mathrm{tr}(\mathbf{W}^\top \mathbf{S_B} \mathbf{W}) \quad \text{s.t.} \quad \mathrm{tr}\,(\mathbf{W}^\top \mathbf{S_W} \mathbf{W}) = \mathbf{I} \quad (4)$$

2) Domain Adaptation Linear Discriminant Analysis

Vidovic et al. [1] introduced a new approach for the transfer learning of LDA models. Statistical attributes such as mean, and covariance estimated solely from the calibration dataset may not be reliable due to insufficient quantity. Transfer learning based on LDA shrinks class-wise mean and covariance matrix toward a more abundant training dataset. For class j, the calculations for the shrunk class-wise covariance and mean are:

$$\widetilde{\boldsymbol{\mu}_j} = (1-\tau)\boldsymbol{\mu}_j^{train} + \tau \boldsymbol{\mu}_j^{calib} \quad (5)$$

$$\widetilde{\mathbf{S}_j} = (1-\lambda)\mathbf{S}_j^{train} + \lambda \mathbf{S}_j^{calib} \quad (6)$$

B. Quantification of Covariate Shift

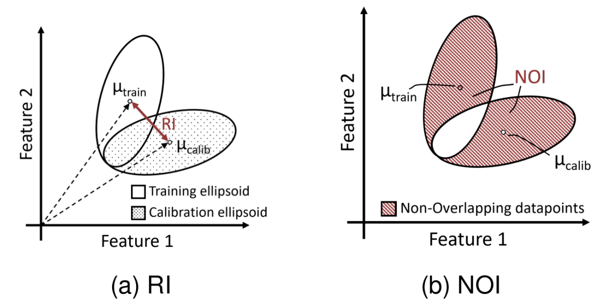

Fig. 1. Illustrations of (a) Repeatability Index (RI) and (b) Non-overlapping index (NOI)

1) Repeatability Index

RI initially introduced by Bunderson and Kuiken [8], serves as a metric to quantify the separation between two clusters (Fig. 1a). In contrast to the original formulation, which calculates the average Mahalanobis distance across multiple gestures, our research focuses specifically on individual gestures. Therefore, our approach to calculating the RI differs from the original method. We define μ_{train} and S_{train} as the mean and covariance of the cluster for training data, and μ_{calib} as the mean of the cluster for calibration data. RI is calculated as follows:

$$\mathrm{RI} = \frac{1}{2}\sqrt{(\mu_{train} - \mu_{calib})^T \mathbf{S}_{train}^{-1} (\mu_{train} - \mu_{calib})} \quad (7)$$

The decision to use the Mahalanobis distance instead of the more straightforward Euclidean distance is primarily due to the different units of different features.

2) Non-overlapping Index

RI can be challenging to interpret due to the arbitrary scaling of feature dimensions and the variance within gesture clusters, which complicates the accurate interpretation of the distance between two clusters. To enhance understanding of the positioning of gesture clusters in feature space, we proposed the Non-Overlapping Index (NOI), which quantifies the degree of non-overlapping between the training and calibration ellipsoids (Fig. 1b). To calculate NOI, a binary classifier is implemented to determine whether a given sample is from the training cluster or the calibration cluster. NOI is determined by the classification rate:

$$\mathrm{NOI} = \frac{1}{n_{feats}} \sum_{k=1}^{n_{feats}} \left(\frac{\text{Correctly Classified}}{\text{Total Classified}} \times 100\%\right) \quad (8)$$

Here, n_{feats} denotes number of features. An NOI value of 100% indicates perfect separability, with no overlap across clusters, whereas a 0% NOI indicates the two clusters are linearly indistinguishable. The primary advantage of the NOI metric is its capacity to verify the adjacency of two clusters when NOI is smaller than 100%. LDA is selected for binary classification due to its minimal computation.

3) Non-overlapping Cluster Count

In addition to NOI, we have also proposed the Non-overlapping Cluster Count (NCC). NCC counts instances where the training cluster does not overlap with the calibration cluster in n-fold cross-validation evaluation. Furthermore, to enable comparisons across multiple datasets, we report NCC in terms of percentage of the number of overlapped instances over the total number of cross-validation iterations. The threshold to determine non-overlapping instances is established at NOI $>$ 99.5%.

C. Calibration of sEMG Gesture Recognition

1) Baseline Models

In this investigation, three baseline models were selected for benchmarking purposes; each has different data handling and model training approaches.

- CALIB ONLY: In the baseline model, we trained the LDA exclusively on the calibration dataset, which consisted of only one single repetition of each gesture. This method serves as a strong comparative baseline. The successful leverage of previously recorded data is determined by whether the classification approach enhances performance over this baseline.
- LDA Data-Aggregation (LDA AGG): Here, we combined the calibration dataset with all previously recorded data into a single dataset to train LDA classifier without considering the difference in recorded conditions.
- LDA Domain Adaptation (LDA DA): We merged all previously recorded data into a single training dataset. Then, the domain adaptation technique for LDA [1] was applied to this combined training dataset and the calibration dataset.

2) LDA Domain Adaptation with Minimum RI Sample Selection

In contrast to baseline methods of LDA AGG and LDA DA, which indiscriminately group all pre-recorded data, here we propose a more strategic approach for sample inclusion based on the RI metric. For each gesture in the calibration set, we selected the full set of gesture repetitions that were recorded under different conditions that achieved the lowest RI score. The inclusion of the entire set is important to account for the variation in gesture repetition. The selected set for each gesture can originate from different conditions. The details of our method are as follows:

- RI AGG: This method extends LDA AGG by combining data from the calibration dataset with training set selected based on minimum RI score.
- RI DA: Here, LDA DA was applied to the calibration dataset and the training set that had the smallest RI score. This approach examined whether limiting the training set to those closest in Mahalanobis distance to the calibration set reduces recognition errors.
- MAX RI DA: In this approach, LDA DA was applied to the training set selected based on the maximum RI score. This method was included to investigate the impact of RI distance on LDA DA.

3) Hyper-parameter Tuning

To determine the optimal values of τ and λ of LDA DA, we utilized grid search approach by varying these hyperparameters from 0 to 1 in steps of 0.2. To mitigate information leakage – a scenario where test data is contaminated into training phase resulting in misleading outcomes, we implemented two approaches for tuning the hyperparameters τ and λ:

- *Cross-Subject*: For subject i, the hyperparameters τ and λ were chosen based on the combination that achieved the lowest test error for all other subjects $j \neq i$.
- *Val-Error*: This approach involved randomly reserving 20% of the calibration dataset for validation purposes. The hyperparameters τ and λ are then selected based on the combination that yielded the lowest error rate in this validation subset.

III. Experiment Settings

Fig. 2. Depiction of five distinct arm positions (P1-P5) from the dataset by Khushaba et al. [9]

A. Khushaba Multiple Limb Position Dataset

The Khushaba et al. [9] Multiple Limb Position dataset contains EMG recordings from 11 subjects. The dataset consists of eight gestures (wrist flexion, wrist extension, pronation, supination, power grip, pinch grip, open hand, and rest) across five limb positions (P1 to P5, Fig. 2). Each gesture was performed six times by the participants. Seven channels of EMG signals were recorded using the Delsys DE 2.x system at a 4000Hz sampling rate. The EMG data was preprocessed with a bandpass filter with a 20–450 Hz cutoff frequency and a 50 Hz notch filter for noise removal. We further down sampled EMG signal to 2000 Hz. We extracted features using Time Domain 5 (TD5) feature set [12] with a window size of 250 ms and a step size of 50 ms.

We quantified covariate shift in the dataset under two scenarios: intra-conditions and inter-conditions. The intra-condition scenario refers to cases within the same limb position, while the inter-condition scenario refers to covariate shifts occurring between two different limb positions. The training dataset comprised of five repetitions, and the calibration set contained a single repetition distinct from those in the training set. Six-fold cross-validation was conducted for both scenarios.

In our domain adaptation experiment, we designated one limb position as a novel condition for calibration. With this novel limb position, one repetition of each gesture was used to form the calibration dataset, while the remaining repetitions were used for testing. For other limb positions, we included all six repetitions of each gesture to form the data training dataset. We employed six-fold cross-validation on the dataset to accurately evaluate our methods.

B. Multi-Day Ninapro DB6 Dataset

In the Ninapro Db6 dataset [10], sEMG data was collected from 10 participants across 10 recording sessions spanning 5 days. During each session, participants were instructed to perform seven distinct grasping motions: large diameter, adducted thumb, index finger extension, medium wrap, writing tripod, power sphere, and precision sphere. Each grasp was repeated on two different objects, alternating between odd and even indexed repetitions. Each combination of grasp type and object was repeated six times. The dataset was recorded using 14 Delsys Trigno EMG electrodes and sampled at 2 kHz. The sEMG signals were notch-filtered at 50 Hz to remove powerline interference. We further applied a 4th-order Butterworth filter with a cutoff frequency of 20–450 Hz. Moreover, we removed the transient of gesture execution and only kept the central 1000 ms. We extracted TD5 features from the filtered sEMG by applying moving window with window size of 250 ms and step size of 50 ms. Additionally, all our experiments focused solely on even-indexed repetitions.

For quantifying covariate shifts, we defined intra-conditions shifts as shifts that occur within the same session, whereas inter-condition shifts refer to covariate shifts between two different sessions that were recorded with at least one night in between. We utilized five repetitions of each gesture for our training dataset and one repetition for the calibration dataset. Additionally, we conducted a six-fold cross-validation to ensure comprehensive evaluation.

For domain adaptation experiment, we selected one session ranging from D3_T1 to D5_T2 as a novel condition. In this selected novel session, one repetition was designated as the calibration dataset, while the remaining repetitions formed the test dataset. All gesture repetitions from earlier sessions, which were recorded at least one night apart, were compiled to create the training dataset. To achieve accurate evaluation, six-fold cross-validation was conducted for each novel session throughout the experiment.

IV. Results

A. Quantification of Covariate Shift in Intra-Conditions and Inter-Conditions Scenarios

Figure 3 illustrates quantification result of covariate shift between the Ninapro DB6 and the Multiple Limb Position datasets on four key metrics: recognition error, RI, NOI, and NCC. Notably, the Ninapro DB6 dataset exhibited substantially higher recognition error in both intra-condition and inter-condition scenarios compared to the Multiple Limb Position dataset (Fig. 3a). For the Ninapro DB6, the recognition error was recorded at 26.453% for intra-condition scenarios and increased to 58.482% for inter-condition scenarios. In contrast, the Multiple Limb Position dataset exhibited a lower intra-condition error of 9.181% and inter-condition error of 36.478%. A similar trend is also evident in the RI metric (Fig. 3b), where distance between gesture clusters was significantly higher in Ninapro DB6 for both intra- and inter-condition.

The evaluation of the non-overlapping metrics of NOI and NCC revealed several intriguing aspects of the two datasets. On the Multiple Limb Position dataset, it was observed that the NOI score in the intra-condition scenario, despite being the lowest, was still considerably high at 85.319% (Fig. 3c). In contrast, the NOI scores for the Multiple Limb Position in intra-conditions scenario and for the Ninapro DB6 dataset in both intra- and inter-conditions scenarios were substantially higher, all exceeded 98%. Regarding the NCC score (Fig. 3d), it was observed that for Multi Limb Dataset, the occurrence of non-overlapping instances was low at 2.235% for intra-condition scenario. In contrast, the Ninapro DB6 dataset exhibited a much higher NCC score in the intra-condition scenarios, reaching 44.643%. For inter-condition scenarios, the NCC scores were 63.826% for the Multiple Limb Position dataset and an even higher 96.22% for the Ninapro DB6 dataset.

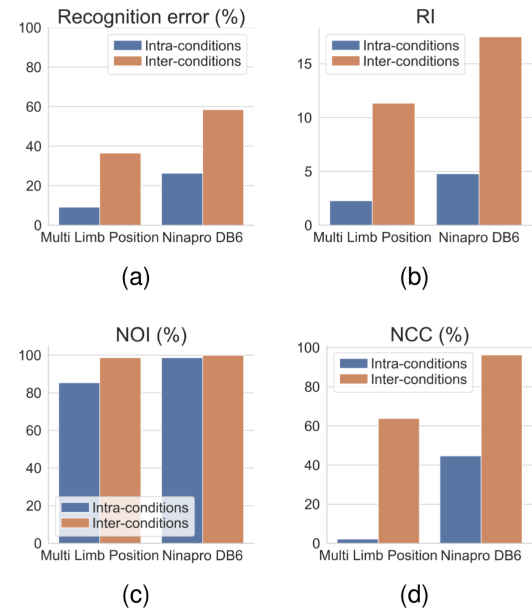

Fig. 3. Quantification of intra- and inter-conditions covariate shift for two datasets in (a) Recognition Error (b) RI (c) NOI (d) NCC

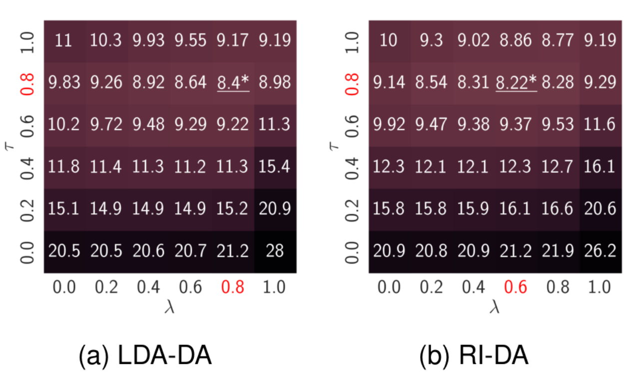

Fig. 4. The effect of τ and λ hyperparameter on recognition errors of (a) LDA-DA (b) RI-DA for Multiple Limb Position Dataset.

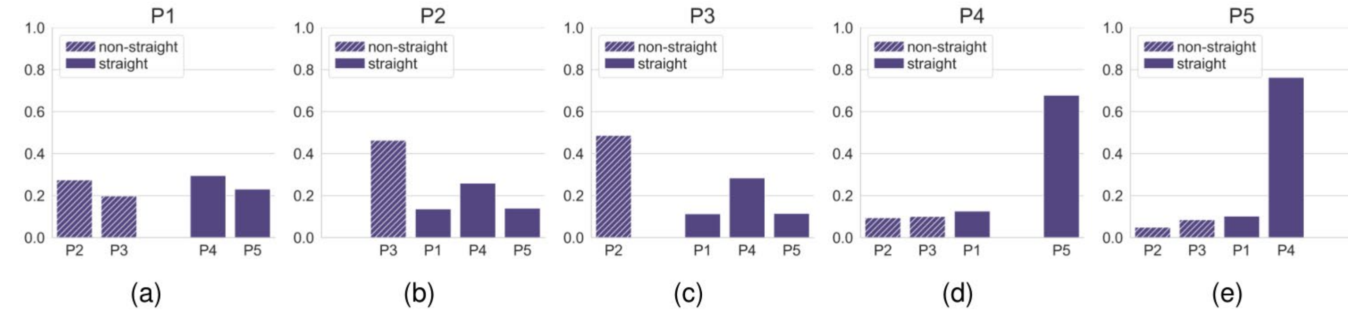

Fig. 5. Frequencies of selected sources for domain adaptation to novel limb positions with minimum RI (RI DA) of (a-e) P1-P5

B. Domain Adaptation for Multiple Limb Position Dataset

TABLE I
RESULTS OF HYPERPARAMETER TUNING STRATEGIES FOR MULTIPLE LIMB POSITION DATASET.

	cross subject	val error
LDA DA	9.531% ± 4.323	**8.782% ± 3.994**
RI DA	9.447% ± 4.406	**8.469% ± 4.033**

Fig. 4 depicts the effect of the hyperparameters τ and λ on recognition errors of LDA DA and RI DA for Multiple Limb Position dataset. The optimal hyperparameters for LDA-DA were identified as (τ=0.8, λ=0.8) and for RI-DA as (τ=0.8, λ=0.6), with RI-DA showing a lower recognition error at 8.223% compared to LDA-DA's figure at 8.401%. However, as these results stem from a retrospective analysis with data leakage, they may not accurately represent true performance. Table I illustrates the results of more realistic hyperparameter tuning procedures of *cross subject* and using *validation error*. We found that validation-error-based selection of τ and λ can reduce recognition error by at least 0.75%.

Table II compares the recognition rates of six different classification approaches for Multiple Limb Position dataset. Notably, the CALIB ONLY method established a strong baseline, averaging an error rate of 9.181% across all limb positions. The DATA AGG approach, which aggregates data from all positions, resulted in an increased error rate of 18.232%. The more selective RI AGG strategy led to a reduced error rate of 13.545%, although this was still higher than the baseline set by CALIB ONLY. Both LDA DA and RI DA effectively lowered the recognition error compared to CALIB ONLY, with LDA DA achieving an error rate of 8.782%, and RI DA recording the lowest error at 8.469%. Despite unfavorable cluster distance conditions MAX RI DA maintained performance compared with the baseline of CALIB ONLY, with the error rate slightly increased to 9.24%.

To further understand the effect of limb position, we analyzed the frequency of limb positions selected for RI DA. Fig. 5 illustrates the frequency of the selected limb position origin. We divided the five limb positions into non-straight elbow (P2 and P3) and straight elbow positions (P1, P4, and P5). It was observed that P2 and P3 frequently selected each other based on the minimum RI score. A similar pattern was

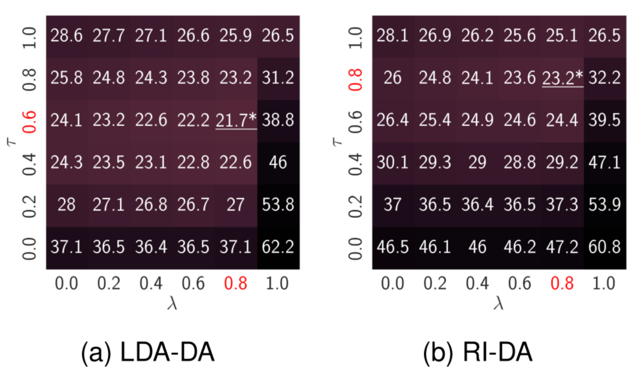

Fig. 6. The effect of τ and λ hyperparameters on recognition errors of (a) LDA-DA (b) RI-DA of the Ninapro DB6 dataset.

TABLE II
RECOGNITION ERRORS FOR THE MULTIPLE LIMB POSITION DATASET.

	CALIB ONLY	DATA AGG	RI AGG	LDA DA	MAX RI DA	RI DA
P1	5.232% ± 3.859	18.469% ± 11.137	12.508% ± 7.444	5.245% ± 3.312	5.648% ± 3.84	**5.108% ± 3.705**
P2	10.739% ± 4.639	19.89% ± 8.537	14.832% ± 6.192	10.287% ± 4.509	10.265% ± 4.687	**9.95% ± 4.62**
P3	12.263% ± 8.065	24.736% ± 10.239	17.615% ± 7.768	12.482% ± 8.355	12.715% ± 8.437	**12.047% ± 7.689**
P4	8.826% ± 5.451	11.434% ± 6.706	10.342% ± 6.55	7.926% ± 4.72	8.927% ± 5.201	**7.662% ± 4.179**
P5	8.845% ± 4.892	16.631% ± 10.084	12.429% ± 8.16	7.97% ± 4.939	8.644% ± 5.059	**7.578% ± 4.616**
ALL	9.181% ± 4.427	18.232% ± 7.118	13.545% ± 5.565	8.782% ± 3.993	9.24% ± 4.197	**8.469% ± 4.033**

also evident with P4 and P5. Interestingly, for limb position P1, the selection frequency was relatively even between straight and non-straight positions.

C. Domain Adaptation on Multi-Day Ninapro DB6 Dataset

Fig. 6 depicts the effect of hyper-parameters τ and λ on the recognition error for the Ninapro Db6 dataset of both LDA DA and RI DA. The optimal (τ, λ) pair for LDA DA was (0.6, 0.8), corresponding to a recognition error of 21.721%, while for RI DA, it was (0.8, 0.8) corresponding to error of 23.216%. Under realistic tuning using *validation error*, as shown in Table III, both LDA DA and RI DA achieved a similar recognition error of approximately 23.4%. Notably, RI DA experienced a substantial reduction in the recognition error, decreasing by 3.5%, compared to using *cross-subject* hyper-parameter tuning.

TABLE III
RESULTS OF HYPERPARAMETER TUNING STRATEGIES FOR NINAPRO DB6.

	cross subject	**val error**
LDA DA	23.373% ± 10.615	**23.35% ± 11.504**
RI DA	26.973% ± 11.543	**23.383% ± 11.568**

Table IV presents the recognition error rates for six different classification methods for the Ninapro DB6 dataset. The CALIB ONLY baseline achieved a recognition error of 26.453%, which is notably lower than 31.053% in error rate of the DATA AGG method and 28.55% in error rate of the RI AGG approach. Both LDA DA and RI DA showed similar performance, with the error rates of 23.35% and 23.385%, respectively, 3% reduction from the CALIB ONLY baseline. Additionally, MAX RI DA also recorded a lower recognition error compared to CALIB ONLY, at 25.154%.

V. DISCUSSIONS

With reference to Rayleigh-Ritz quotient (Eq.1), for sEMG gesture recognition, between-class variance is influenced by sensor placement, window size [13], selection of gesture and feature sets [14], and recording conditions [15]. On the other hand, within-class variance is primarily influenced by user expertise [8]. In our study, we focused specifically on analyzing the within-class characteristics. Additionally, we conducted a comprehensive analysis of two distinct sEMG datasets, which differ in terms of the number of electrodes, their placement, and the selected gesture sets. The Ninapro DB6 dataset mainly comprises similar grasping gestures, unlike the Multiple Limb Position dataset, which features a more diverse gesture set. This complexity of gesture sets was reflected in the intra-condition recognition error and RI score (Fig. 3a and Fig. 3b).

NOI score (Fig. 3c) reveals a high level of non-overlap, 85.319%, in the intra-condition of the Multiple Limb Position dataset. We attribute this distinctiveness more to sEMG instability than to user expertise. The NCC score in the inter-condition scenario of the Ninapro DB6 dataset is significantly high at 96.22%, while the Multiple Limb Position dataset shows a more moderate figure at 63.836% (Fig. 3d). This difference suggests that the covariate shift in the cross-day scenario of Ninapro DB6 is more pronounced than the shift across limb positions in the Multiple Limb Position dataset.

Throughout our experiments with domain adaptation for the two datasets, we made several key observations. Using only the calibration data to train LDA model (CALIB ONLY) provided a strong baseline. It is especially noticeable in the Multiple Limb Position dataset, where the recognition error rate was as low as 9.181%. In contrast, indiscriminately combining data from all conditions with the calibration dataset (DATA AGG) to train classifiers resulted in increased recognition error ranging from 5% to 9%, compared to the baseline of CALIB ONLY on both datasets.

Domain adaptation strategies of LDA DA and our proposed RI DA method demonstrated a consistent reduction in the recognition error compared with the baseline of CALIB ONLY for both datasets. The amount of reduction ranged from 0.4% for the Multiple Limb Position dataset to 3.1% for the Ninapro DB6. Additionally, for the Multiple Limb Position dataset, our proposed RI DA method performed better than LDA DA by 0.313%. A minimal increase between RI DA and LDA DA of 0.033% was observed for Ninapro DB6. Moreover, LDA DA and RI DA optimal hyper-parameters, τ and λ, can be adaptively selected based on the leave-out validation dataset (Table I and III). Additionally, the RI DA method was trained on a significantly reduced dataset size compared to LDA DA. This indicates that the majority of the collected data may not be

TABLE IV
RECOGNITION ERRORS FOR NINAPRO DB6 .

	CALIB ONLY	**DATA AGG**	**RI AGG**	**LDA DA**	**MAX RI DA**	**RI DA**
D3_T1	25.968% ± 11.503	30.799% ± 8.415	31.077% ± 13.546	**21.935% ± 10.954**	23.736% ± 10.012	22.808% ± 12.632
D3_T2	27.175% ± 14.732	30.439% ± 12.387	28.02% ± 13.295	24.896% ± 14.262	27.451% ± 15.279	**24.854% ± 13.625**
D4_T1	25.144% ± 10.689	25.075% ± 9.278	24.361% ± 10.084	21.702% ± 10.895	22.078% ± 9.573	**21.498% ± 10.202**
D4_T2	26.112% ± 9.519	31.865% ± 15.049	28.011% ± 11.808	23.85% ± 10.679	25.564% ± 10.388	**23.29% ± 10.663**
D5_T1	27.538% ± 15.748	29.612% ± 17.914	30.226% ± 15.91	**23.568% ± 15.58**	26.197% ± 14.459	23.837% ± 15.331
D5_T2	26.782% ± 14.745	38.527% ± 18.614	29.607% ± 14.695	24.151% ± 14.894	25.896% ± 13.916	**24.009% ± 14.584**
ALL	26.453% ± 11.606	31.053% ± 9.638	28.55% ± 11.818	**23.35% ± 11.504**	25.154% ± 10.809	23.383% ± 11.568

essential for effective sEMG calibration. Notably, under suboptimal conditions of MAX RI DA, it still obtains improved performance for Ninapro DB6 and a small decrease of 0.059% for the Multiple Limb Position dataset over the CALIB ONLY baseline. This outcome highlights the robustness of LDA domain adaptation in general and our adaptive hyper-parameter tuning approach, especially under the substantial covariate shifts observed in the Ninapro DB6 dataset.

In the analysis of the Multiple Limb Position dataset, we observed distinct preferences in the selection between the pairs (P2, P3) and (P4, P5), with P1 was seen as an intermediate between these two pairs (Fig. 5). For the Ninapro DB6 dataset, despite the later sessions having a larger pool of gesture executions, no discernible trend was observed between D3 and D5. This absence of a discernible pattern in the Ninapro DB6 dataset can be largely attributed to the less controlled placement of electrodes. The simultaneous effects of electrode shift and changes in user behavior make it challenging to identify the exact cause of performance stagnation.

VI. Conclusions

Domain adaptation is an effective technique for sEMG gesture recognition performance amid covariate shifts caused by the non-stationary nature of sEMG signals. In this paper, we first quantified the amount of covariate shift under both inter- and intra-condition scenarios for two public datasets. Furthermore, we addressed the limitations of the RI metric by introducing two new, more intuitive metrics of NOI and NCC that effectively measured the degree of nonoverlap between gesture clusters. Specifically, using the NCC metric, we found that the covariate shift in the DB6 dataset over multiple days was more pronounced than across limb position for the Multiple Limb Position dataset. Secondly, we extended domain adaptation in LDA using an RI-based sample selection strategy, achieving a performance improvement of up to 3% over models trained only with calibration data. Compared to LDA domain adaptation without sample selection, our proposed RI DA approach reduced the recognition error by 0.3% for the Multiple Limb Position dataset, while maintaining similar performance for the Ninapro DB6 dataset. Moreover, the RI DA method demonstrated that effective LDA domain adaptation can be achieved using a smaller dataset, suggesting that a significant portion of the data is not essential for the domain adaptation process. Our future research aims to improve data recording procedures and domain adaptation to further reduce training burden on users and better utilize data recorded under various conditions [16].

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